



Rossmoyne Senior High School

Semester One Examination, 2017

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section One:
Calculator-free

SOLUTIONS

Student Number: In figures

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In words

Your name

Time allowed for this section

Reading time before commencing work: five minutes

Working time: fifty minutes

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	51	36
Section Two: Calculator-assumed	13	13	100	90	64
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet.
3. You must be careful to confine your response to the specific question asked and to follow any instructions that are specified to a particular question.
4. Additional working space pages at the end of this Question/Answer booklet are for planning or continuing an answer. If you use these pages, indicate at the original answer, the page number it is planned/continued on and write the question number being planned/continued on the additional working space page.
5. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
6. It is recommended that you do not use pencil, except in diagrams.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section One: Calculator-free

35% (51 Marks)

This section has **seven (7)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1

(6 marks)

One Australian dollar can buy 50 Indian rupees.

- (a) How many Indian rupees can be bought for 30 Australian dollars? (1 mark)

Solution
$30 \times 50 = 1\,500$ rupees
Specific behaviours
✓ multiplies correctly

- (b) Determine how many Australian dollars are required to buy 50 000 rupees. (1 mark)

Solution
$\frac{50000}{50} = \$1000$
Specific behaviours
✓ divides correctly

- (c) A tourist going to India wants to take 50 000 rupees with them. Their bank uses the above exchange rate but also adds a commission of 1%. How much will the rupees cost the tourist altogether? (2 marks)

Solution
$1000 \div 100 = \$10$ commission Total cost is $\$1000 + \$10 = \$1010$
Specific behaviours
✓ calculates commission ✓ calculates total cost

- (d) Over the next few months, the exchange rate is expected to change so that one Australian dollar will buy 52 rupees. Determine the percentage increase in the exchange rate from 50 to 52 rupees. (2 marks)

Solution
$\frac{52 - 50}{50} \times 100 = \frac{2}{50} \times 100 = 4\%$
Specific behaviours
✓ calculates change ✓ calculates percentage

Question 2

(8 marks)

(a) Determine the value of

(i) $a^2 + 2a + 3$ when $a = 7$.

(2 marks)

Solution
$49 + 14 + 3 = 66$
Specific behaviours
✓ squares and doubles ✓ sums terms

(ii) $x - xy - y$ when $x = -2$ and $y = 3$.

(2 marks)

Solution
$-2 - (-2 \times 3) - 3 = -2 + 6 - 3 = 1$
Specific behaviours
✓ substitutes correctly ✓ evaluates

(b) Calculate the price-to-earnings ratio for a share with a market price of \$2.75 and annual earnings per share of 25 cents.

(2 marks)

Solution
$PE = \frac{275}{25} = 11$
Specific behaviours
✓ uses PE formula with correct units ✓ evaluates ratio

(c) The price-to-earnings ratio for a share is 20. If the market price of the share is \$1.10, determine the annual earnings per share.

(2 marks)

Solution
$20 = \frac{110}{E} \Rightarrow E = \frac{110}{20} = \frac{55}{10} = 5.5 \text{ cents per share}$
Specific behaviours
✓ uses PE formula correctly ✓ solves for EPS, stating units

Question 3

(10 marks)

Four matrices A, B, C and D are shown below.

$$A = \begin{bmatrix} 0 & 2 & 4 \\ -3 & 1 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 8 \\ 0 \\ -12 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 5 \\ -1 & 2 & 0 \\ 4 & 3 & -2 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 2 \end{bmatrix}.$$

- (a) State the size of the square matrix.

(1 mark)

Solution
3×3
Specific behaviours
✓ states rows and columns

- (b) Calculate $2C + 5I$, where I is the 3×3 identity matrix.

(3 marks)

Solution
$2C + 5I = \begin{bmatrix} 2 & 0 & 10 \\ -2 & 4 & 0 \\ 8 & 6 & -4 \end{bmatrix} + \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 7 & 0 & 10 \\ -2 & 9 & 0 \\ 8 & 6 & 1 \end{bmatrix}$
Specific behaviours
✓ calculates $2C$ ✓ uses $5I$ ✓ calculates correct sum

- (c) Calculate $D \times A$.

(2 marks)

Solution
$DA = \begin{bmatrix} -6 & 12 & 16 \end{bmatrix}$
Specific behaviours
✓ product is 1×3 ✓ correct product

- (d) Determine the values of p and q given that $p \times \begin{bmatrix} -2 \\ q \\ 3 \end{bmatrix} = B$.

(2 marks)

Solution
$p \times -2 = 8 \Rightarrow p = -4$ $-4 \times q = 0 \Rightarrow q = 0$
Specific behaviours
✓ determines p ✓ determines q

- (e) If $E = A \times C$, determine the value of e_{21} .

(2 marks)

Solution
Multiply row 2 of A by column 1 of C to get $E_{2,1} = -3 + -1 + -8 = -12$
Specific behaviours
✓ uses correct row and column ✓ correct result

Question 4**(6 marks)**

- (a) An investor received a dividend of \$240 on 60 shares in a mining company. Calculate the dividend per share. (1 mark)

Solution
$\$240 \div 60 = \4 per share
Specific behaviours
✓ calculates dividend per share

- (b) Another company paid a dividend of 25 cents per share to an investor who held 2 000 shares in the company.

- (i) Calculate the total dividend the investor was paid. (1 mark)

Solution
$0.25 \times 2000 = \$500$
Specific behaviours
✓ calculates total dividend

- (ii) The investor had to pay income tax of 30% on the total dividend. Determine how much the investor received after tax had been deducted. (2 marks)

Solution
$0.3 \times 500 = \$150$ $500 - 150 = \$350$
Specific behaviours
✓ calculates tax ✓ calculates remainder

- (c) A shareholder has 50 000 shares with a market price of 40 cents in a technology company, and the company has announced a dividend per share of 1.5% of the share price. Determine the total dividend the investor can expect. (2 marks)

Solution
$50000 \times 0.4 \times 0.015 = 200 \times 0.015 = \300
Specific behaviours
✓ uses correct method ✓ determines correct dividend, with units

Question 5

(7 marks)

A freight company has a display at a trade exhibition that includes various scale models of its standard shipping containers (such a container can be closely modelled by a rectangular prism). A full-size standard container is 12.0 m long, 2.4 m wide and 2.5 m high.

- (a) One of the scale models was built to a one-tenth scale of a full-size container. State the length, width and height of this model, in centimetres. **(3 marks)**

Solution		
$L = \frac{12}{10} \times 100 = 120,$	$W = \frac{2.4}{10} \times 100 = 24,$	$H = \frac{2.5}{10} \times 100 = 25 \text{ cm}$
Specific behaviours		
✓ uses scale factor ✓ converts from m to cm ✓ clearly states length, width, height		

- (b) Another model was made to a scale of 1:100. If the area of the floor of a full-size container is 288 000 cm², determine the area of the floor of this model. **(2 marks)**

Solution
$A = \frac{288000}{100 \times 100} = 28.8 \text{ cm}^2$
Specific behaviours
✓ uses square of scale factor ✓ determines area

- (c) The largest model was built to a one-half scale of a full-size container. Determine the volume of this scale model. **(2 marks)**

Solution
$V = \frac{2.4}{2} \times \frac{2.5}{2} \times \frac{12}{2} = \frac{2.4}{8} \times 30 = 0.3 \times 30 = 9 \text{ m}^3$
Specific behaviours
✓ uses cube of scale factor ✓ determines volume, stating units * <i>challenging calculation</i>

Question 6**(6 marks)**

Ann and Ben held 400 and 350 shares respectively in three public companies: Ria, Skip and Tap.

Ann held 200 in Ria, 50 in Skip and the remainder in Tap. Ben held 50 in Tap, twice that number in Ria and the remainder in Skip.

- (a) Display this information in M , a 2×3 matrix.

(2 marks)

Solution			
	R	S	T
A	200	50	150
B	100	200	50
Specific behaviours			
✓ uses correct size and labels			
✓ correct entries			

- (b) The market value of Ria, Skip and Tap shares is \$1.20, \$3.50 and \$2.10 respectively. Display this information in matrix N , so that matrix $M \times N$ will show the values of each person's share holdings. **Do not** calculate $M \times N$.

(1 mark)

Solution	
$N =$	$\begin{bmatrix} 1.20 \\ 3.50 \\ 2.10 \end{bmatrix}$
Specific behaviours	
✓ correct column matrix	

- (c) Explain what information the result of $[1 \ 1] \times M$ will show.

(1 mark)

Solution	
The total number of shares held by the two, in each of Ria, Skip and Tap.	
Specific behaviours	
✓ clear explanation	

- (d) A stock market analyst expects the values of the shares in six months' time to be found from the result of $\begin{bmatrix} 1.5 & 0 & 0 \\ 0 & 1.2 & 0 \\ 0 & 0 & 1.1 \end{bmatrix} \times N$. Use this information to calculate the expected price of Skip shares.

(2 marks)

Solution	
Skip shares will increase by 20%.	
Expected price: $\$3.50 + 0.35 + 0.35 = \4.20	
Specific behaviours	
✓ evidence of 20% increase	
✓ correct new price	

Question 7

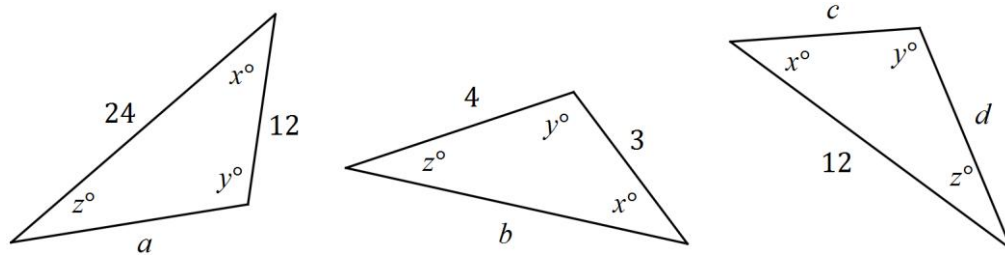
(8 marks)

- (a) Write the scale 5 mm : 30 cm as a ratio in simplest form and without units.

(2 marks)

Solution
$5 \text{ mm} : 300 \text{ mm} = 1 : 60$
Specific behaviours
✓ converts measurements to same unit ✓ simplifies ratio

- (b) Three similar triangles are shown below, although they are not drawn to the same scale. Some of the side lengths, in centimetres, are shown.



Determine the lengths a , b , c and d .

(4 marks)

Solution
Lengths of LH triangle are $\frac{12}{3} = 4 \times$ middle triangle and $\frac{24}{12} = 2 \times$ RH triangle. $a = 4 \times 4 = 16 \text{ cm}$, $b = \frac{24}{4} = 6 \text{ cm}$, $c = \frac{12}{2} = 6 \text{ cm}$ and $d = \frac{16}{2} = 8 \text{ cm}$.
Specific behaviours
✓ uses ratio of lengths of LH and middle triangles ✓ determines a and b ✓ uses ratios of lengths of RH and another triangle ✓ determines c and d

- (c) A student used a map drawn to a scale of 1: 10 000 to plan a walk. The length of the walk on the map was 5.5 cm. State, with reasoning, whether the length of the walk was more than a kilometre.

(2 marks)

Solution
No, walk was less than 1 km.
$\frac{5.5 \times 10000}{100} = 550 \text{ m}$
Specific behaviours
✓ states less ✓ states walk is 550 m long.

Additional working space

Question number: _____

Additional working space

Question number: _____

